مدارس الإمارات الوطنية
Emirates National Schools

Grade: 5 | Term 1, Week 10 (2 Nov to 6 Nov) | TCD: Future Skills 5: Prototyping and Smart Making | Session length: 1 hr 30 min

T1 WEEK: 10	**Lesson 1: 3D Modelling a Car Logo**
Standard(s):	Math: 5.G.B.3 and 5.G.B.4 (classify shapes by properties); Visual Arts: analyse form and symmetry; Technology: CAD modelling
Literacy Strategies:	Shape-spotting sheet with the words circle, ring, star, triangle, oval and letter; sentence frame: This logo is made of ...
Assessment:	Shape breakdown sheet; Tinkercad model checked against the reference; live quiz on the TCD website.
WALT (Learning Target):	We are learning to break a logo into simple shapes and rebuild it as a 3D model.
WILF (Success Criteria):	I can find the basic shapes inside a car logo. I can use rotate, duplicate and mirror to build symmetrical parts. I can use holes to cut out inner shapes. I can scale my model to 60 mm and give it a 4 mm base.
Targeted Vocabulary	logo, symmetry, mirror, duplicate, rotate, scale, ring, reference
Resources	Computers or iPads with Tinkercad, Car logo reference sheet (teacher chooses 6 logos built from simple shapes), Shape-spotting sheet, Tracing paper, Rulers Accompanying resource: Grade5_Week10_Presentation.pptx (slides, hook video and live quiz code on the TCD website), Grade5_Week10_Worksheet and Grade5_Week10_Lesson_Plan.
National Identity Integration	Discuss the UAE's car culture, from the Dubai Police supercar fleet to the Etihad Rail and electric vehicles, and how strong logos help companies become known around the world.
AI/Digital Learning	Tinkercad; printed logo reference sheet (teacher prepared); hook video; live quiz on the TCD website.
Direct Instruction: I Do	Model breaking a logo into shapes (slide 6). Model a ring cut-out and overlapping rings (slide 7). Model duplicate, rotate and mirror (slide 8) and scaling (slide 9).
Guided Practice: We Do	Class breaks down one logo together on the board. Students build one ring together, step by step (slide 10).
Independent Practice: You Do	Students model their chosen logo (slide 12). Challenge levels on the board (slide 13). Live quiz on tablets (slide 14).
Differentiated Instruction: Beginning / On / Above	**Beginning: Choose a logo made of rings or circles and follow the ring step card.** **On: Choose a logo with a star, triangle or letter and use rotate or mirror.** **Above: Choose a complex logo, add a raised border and explain how you used symmetry.**
Closure/Reflection	Side-by-side check: model next to the reference. One fix for next week. Exit ticket (slide 15). Save and name the file Grade5_CarLogo_YourName.
Inclusion / EAL Support	☑ Visual Supports ☑ Chunked Instructions ☐ Reduced Language Load ☑ Model / Worked Example ☐ Pre-teach Vocabulary ☐ TA/LSA Support ☐ Other: ☐ Extra Processing Time ☐ Text Read by TA/LSA ☑ Flexible Grouping ☐ Assistive Technology ☐ Movement / Sensory Break ☐ Scaffold / Sentence Starters

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G&T / Extension	☑ Extension / Challenge Task ☐ Real-World Application ☑ Higher-Order Questioning ☐ Curriculum Compacting ☑ Depth & Complexity Task ☐ Above-Grade-Level Texts ☐ Independent Inquiry ☐ Student-Led /Passion project ☐ Leadership / Peer Coaching ☐ Open-Ended Task ☐ Other: _____